Economic Resilience Under Disruption: A DIIM-Based Framework for Key Sector Identification in Singapore

Yeoh Yu Yong

SCHOOL OF PHYSICAL AND MATHEMATICAL SCIENCES

A DISSERTATION SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN MATHEMATICAL SCIENCES

2026

Contents

Abstract ii

Acknowledgement iii

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 The Dynamic Inoperability Input-Output Model . . . 1
- 1.3 The Key Sector Identification Problem . . . 1
- 1.4 Objectives . . . 2
- 1.5 Report Organisation . . . 2

2 Methodology 3
- 2.1 The Dynamic Inoperability Input-Output Model . . . 3
  - 2.1.1 Inoperability . . . 3
  - 2.1.2 From Leontief to IIM . . . 3
  - 2.1.3 The DIIM . . . 4
  - 2.1.4 Initial Inoperability . . . 5
  - 2.1.5 Elasticity Coefficient . . . 5
  - 2.1.6 Economic Loss . . . 5
- 2.2 Key Sector Identification Methods . . . 5
  - 2.2.1 Intervention and Loss Reduction Evaluation . . . 6
  - 2.2.2 DIIM-Based Identification (Benchmark) . . . 6
  - 2.2.3 Total Output (x_i) . . . 7
  - 2.2.4 PCA $\times x_i$ . . . 7
  - 2.2.5 PageRank $\times x_i$ . . . 7
  - 2.2.6 Backward Linkage $\times x_i$ (BL $\times x_i$) . . . 7
  - 2.2.7 Forward Linkage $\times x_i$ (FL $\times x_i$) . . . 7
- 2.3 Performance Metric . . . 8

3 Case Study 1: COVID-19 Pandemic 9
- 3.1 Data and Model Parameters . . . 9
  - 3.1.1 Input-Output Table . . . 9
  - 3.1.2 Initial Inoperability . . . 9
  - 3.1.3 Model Parameters . . . 9
- 3.2 Results . . . 10
  - 3.2.1 Inoperability Evolution . . . 10
  - 3.2.2 Economic Loss Evolution . . . 10
  - 3.2.3 Identifying Vulnerable Sectors . . . 10

4 Case Study 2: Manpower Disruption **13**
4.1 Background: Singapore's Foreign Worker Dependence . . . 13
4.2 Data and Model Parameters . . . 13
4.2.1 Input-Output Table . . . 13
4.2.2 Initial Inoperability . . . 13
4.2.3 Model Parameters . . . 14
4.3 Results . . . 14
4.3.1 Inoperability Evolution . . . 14
4.3.2 Economic Loss Evolution . . . 14
4.4 Comparison with COVID-19 . . . 15

5 Key Sector Identification Methods **16**
5.1 PCA Applied to the IO Matrix . . . 16
5.2 Simplified Method Rankings . . . 17
5.3 Comparison at Original Disruption Data . . . 17
5.4 Effect of the Number of Key Sectors . . . 18

6 Monte Carlo Robustness Analysis **20**
6.1 Framework . . . 20
6.1.1 Motivation . . . 20
6.1.2 Performance Metric . . . 20
6.2 Monte Carlo Design . . . 20
6.2.1 Perturbation of Real $q(0)$ . . . 20
6.2.2 Simulation Procedure . . . 21
6.3 Results: Performance Across k . . . 21
6.3.1 Mean Performance Ratio . . . 21
6.3.2 Close Rates . . . 21
6.3.3 Three Regimes of k . . . 22
6.4 Key Observations . . . 22
6.5 Practical Implications . . . 23

7 Discussion **24**
7.1 The Three Regimes of Intervention Scale . . . 24
7.2 Why Economic Size Dominates Structure . . . 24
7.3 When Do Structural Methods Add Value? . . . 25
7.4 Cross-Scenario Comparison . . . 25
7.5 National Security Implications . . . 25
7.6 Limitations . . . 26
7.7 Future Work . . . 26

8 Conclusion **27**

Appendix A: COVID-19 Scenario Data **31**

Appendix B: Manpower Scenario Data **33**

Appendix C: Code **34**

Abstract

This project develops a quantitative framework for assessing economic resilience under disruption using the Dynamic Inoperability Input-Output Model (DIIM) [1, 2]. The DIIM captures how sectoral disruptions propagate through input-output linkages and how the economy recovers over time. Two disruption scenarios relevant to Singapore are studied: (1) the COVID-19 pandemic, using the 2019 input-output (IO) table aggregated to 15 sectors, and (2) a manpower disruption scenario motivated by potential grey zone conflicts that could trigger large-scale foreign worker departures, using the 2022 IO table with 15 sectors.

In disruption management, a key question is which sectors should receive priority intervention. The DIIM can identify these "key sectors" by simulating the full disruption and ranking sectors by economic loss, but this approach is computationally expensive and requires crisis-time data (initial inoperability $q(0)$, demand perturbation c^*) that may not be available during the early stages of an event. This motivates the search for simplified structural alternatives that use only the IO table.

This study evaluates five simplified methods—Total Output, PCA $\times x_i$, PageRank $\times x_i$, BL $\times x_i$, and FL $\times x_i$—against the DIIM benchmark through Monte Carlo simulation (2 000 random $q(0)$ vectors per scenario). Performance is measured by the ratio of economic loss reduction achieved by each simplified method relative to DIIM. For the 15-sector COVID scenario, the best structural methods (Total Output, PageRank $\times x_i$, BL $\times x_i$, and FL $\times x_i$) match DIIM in 89.8% of cases, and PCA $\times x_i$ matches in 86.6%. For the 15-sector manpower scenario, FL $\times x_i$ and Total Output match in 55.4% of cases, while PCA $\times x_i$ drops to 38.8%. Logistic regression is used to derive decision rules that predict, from observable $q(0)$ features, when each simplified method will perform adequately.

Acknowledgement

I would like to express my sincere gratitude to my supervisor(s) for their guidance, encouragement, and constructive feedback throughout this project. Their expertise and patience were invaluable in shaping the direction of the work and in helping me develop my research and writing skills.

I am also grateful to the members of my committee and the faculty in the department for their insightful comments and support. I would like to acknowledge the technical and administrative staff whose assistance made it possible to carry out the practical aspects of this work.

I would like to thank my colleagues and friends for the discussions, collaboration, and encouragement that made the process both productive and enjoyable. Their willingness to share ideas and offer help at key moments is deeply appreciated.

Finally, I am profoundly thankful to my family for their unwavering support, understanding, and motivation. Their confidence in me has been a constant source of strength throughout my studies.

Chapter 1

Introduction

1.1 Background

Singapore is a small, open, city-state economy whose prosperity depends on deep integration with global supply chains and a large foreign workforce. Approximately one-third of Singapore's labour force consists of non-resident workers, spanning construction, manufacturing, marine shipyard, and services [3]. This dependence, while a source of economic dynamism, is also a structural weakness: any event that rapidly reduces the available workforce—whether a pandemic, a geopolitical crisis, or a grey zone conflict—can propagate through interconnected sectors and inflict cascading economic losses far beyond the initially affected industries.

The COVID-19 pandemic illustrated this. When Singapore imposed the "circuit breaker" lockdown from 7 April to 1 June 2020 [4,5], the disruption was not confined to directly affected sectors. Construction shutdowns deprived building-material manufacturers of demand, transport restrictions disrupted logistics, and hospitality closures cascaded into food supply chains. Understanding these cascading effects requires a framework that captures how sectors depend on each other and how disruptions propagate over time.

Beyond pandemics, Singapore faces national security scenarios that could produce similar or even more severe manpower disruptions. Grey zone conflicts—coercive actions below the threshold of conventional warfare, such as economic coercion, cyber attacks, or hybrid threats—could trigger rapid departures of foreign workers, disrupting sectors that rely heavily on non-resident labour [6]. Unlike COVID-19, where the external demand shock (c^*) was significant, a manpower disruption scenario is characterised by high initial inoperability ($q(0)$) with $c^* = 0$, representing a purely structural disruption that propagates through the production network.

1.2 The Dynamic Inoperability Input-Output Model

The Inoperability Input-Output Model (IIM) and its dynamic extension (DIIM) provide a rigorous framework for modelling such disruptions [1]. Built on the Leontief input-output model, the DIIM measures "inoperability"—the normalised production loss of each sector on a scale from 0 (normal) to 1 (complete shutdown)—and simulates how initial shocks propagate through input-output linkages and how the economy recovers over time. The Shanghai COVID-19 study demonstrated the utility of this approach for quantifying sectoral vulnerability and evaluating risk management strategies [2].

1.3 The Key Sector Identification Problem

One application of the DIIM is identifying "key sectors"—those where targeted intervention (e.g., reducing inoperability by allocating recovery resources) yields the greatest reduction in total economic loss. The DIIM identifies these sectors by running the full dynamic simulation and ranking sectors by cumulative economic loss. However, this approach has two limitations:

1. **Computational cost**: The DIIM requires running the complete simulation for every candidate intervention strategy.

2. **Data requirements**: The DIIM needs crisis-time data—initial inoperability $q(0)$ and the demand perturbation c^*—which may not be available in the early stages of a disruption or during pre-event planning.

This raises the question of whether there exist *simplified structural methods* that can identify key sectors using only the input-output table, which is available in peacetime. Principal Component Analysis (PCA) applied to the integrated IO matrix $H = A(I-A)^{-1}$ is one such method, previously shown to produce sector rankings comparable to DIIM for specific scenarios. However, whether PCA's performance generalises across different disruption profiles (i.e., different $q(0)$ vectors) is an open question.

1.4 Objectives

This study addresses the following objectives:

1. Apply the DIIM framework to two disruption scenarios relevant to Singapore: the COVID-19 pandemic (15 sectors, 2019 IO table) and a manpower disruption (15 sectors, 2022 IO table).

2. Evaluate five simplified structural methods for key sector identification—Total Output, PCA $\times x_i$, PageRank $\times x_i$, BL $\times x_i$, and FL $\times x_i$—against the DIIM benchmark.

3. Determine the boundary conditions (in terms of the initial inoperability vector $q(0)$) under which each simplified method can reliably substitute for the full DIIM simulation.

4. Determine how the performance gap between simplified methods and DIIM varies with the number of sectors targeted for intervention (k), identifying the regimes where simplified methods are sufficient and where DIIM remains essential.

1.5 Report Organisation

Chapter 2 presents the methodology, including the derivation of the DIIM and the formulation of the simplified structural methods. Chapter 3 applies the DIIM to the COVID-19 scenario, presenting results and risk management analysis. Chapter 4 extends the analysis to the manpower disruption scenario. Chapter 5 compares the key sector identification methods across both scenarios. Chapter 6 presents the Monte Carlo robustness analysis across multiple intervention scales. Chapter 7 discusses the implications, and Chapter 8 concludes.

Chapter 2

Methodology

This chapter presents the analytical framework. Section 2.1 derives the DIIM from the Leontief input-output model. Section 2.2 introduces the simplified structural methods for key sector identification. Section 2.3 defines the performance metric used to compare methods.

2.1 The Dynamic Inoperability Input-Output Model

2.1.1 Inoperability

This paper uses the DIIM model to assess the impacts of economic disruptions on the inoperability and economic losses of Singapore's sectors. The DIIM model is derived from the Leontief input-output model and is used to investigate the higher-order transmission effect of input-output linkages between sectors on inoperability. Using the DIIM model requires first introducing the concept of "inoperability", which is defined as follows:

$$\text{Inoperability} = \frac{\text{As Planned Production} - \text{Degraded Production}}{\text{As Planned Production}} \tag{2.1}$$

Where "As Planned Production" represents the output level of a sector under normal production, and "Degraded Production" represents the output level of a sector after being shocked in its production process. "Inoperability" takes values between 0 and 1, with higher values indicating greater damage to production caused by the shock. A value of 1 means the shocked sector has completely lost its production capacity, while a value of 0 means the sector is producing at a normal level.

2.1.2 From Leontief to IIM

Using the concept of inoperability defined above, the IIM model can be derived from the Leontief input-output model [7], as shown in Equation (2.2).

$$x = Ax + c \tag{2.2}$$

Where x is the total output vector; A is the technical coefficient matrix; c is the final demand vector. If we define the output levels and the final demand vector of some shocked sectors as $\tilde{x}$ and $\tilde{c}$ respectively, then we can construct the input-output relationship between sectors after the shock.

Subtracting Equation (2.2) from the shocked-economy formulation:

$$x - \tilde{x} = A(x - \tilde{x}) + (c - \tilde{c}) \tag{2.3}$$

Defining $\hat{x}$ as a diagonalized matrix of the output vector and left multiplying $\hat{x}^{-1}$ on both sides of Equation (2.3):

$$\hat{x}^{-1}(x - \tilde{x}) = \hat{x}^{-1}A(x - \tilde{x}) + \hat{x}^{-1}(c - \tilde{c}) \tag{2.4}$$

Let $q = \hat{x}^{-1}(x - \tilde{x})$, $\quad A^* = \hat{x}^{-1} A \hat{x}$, $\quad c^* = \hat{x}^{-1}(c - \tilde{c})$; then Equation (2.5) can be derived, which is the IIM model.

$$q = A^* q + c^* \tag{2.5}$$

The details of each matrix in the IIM model are shown in Equations (2.6), (2.7), and (2.8):

$$q = \hat{x}^{-1}(x - \tilde{x}) = \begin{bmatrix} \frac{1}{x_1} & 0 & \cdots & \cdots & 0 \\ 0 & \ddots & & & \vdots \\ \vdots & & \frac{1}{x_i} & & \vdots \\ \vdots & & & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} x_1 - \tilde{x}_1 \\ \vdots \\ x_i - \tilde{x}_i \\ \vdots \\ x_n - \tilde{x}_n \end{bmatrix} \tag{2.6}$$

$$A^* = \hat{x}^{-1} A \hat{x} = \begin{bmatrix} a_{11}\frac{x_1}{x_1} & \cdots & a_{1j}\frac{x_j}{x_1} & \cdots & a_{1n}\frac{x_n}{x_1} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1}\frac{x_1}{x_i} & \cdots & a_{ij}\frac{x_j}{x_i} & \cdots & a_{in}\frac{x_n}{x_i} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n1}\frac{x_1}{x_n} & \cdots & a_{nj}\frac{x_j}{x_n} & \cdots & a_{nn}\frac{x_n}{x_n} \end{bmatrix} \tag{2.7}$$

$$\mathbf{c}^* = \hat{\mathbf{x}}^{-1}(\mathbf{c} - \tilde{\mathbf{c}}) = \begin{bmatrix} \frac{1}{x_1} & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \frac{1}{x_i} & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & \frac{1}{x_n} \end{bmatrix} \begin{bmatrix} c_1 - \tilde{c}_1 \\ \vdots \\ c_i - \tilde{c}_i \\ \vdots \\ c_n - \tilde{c}_n \end{bmatrix} \tag{2.8}$$

2.1.3 The DIIM

The IIM model can be extended to the DIIM model by introducing a time variable and an elasticity coefficient matrix describing the recovery capacity of sectors [1]. The discrete form of the DIIM model is:

$$q(t+1) = q(t) + K\left[A^* q(t) + c^*(t) - q(t)\right] \tag{2.9}$$

Where K is the elasticity coefficient matrix describing the recovery capacity of sectors after being shocked in production. Assuming that the recovery capacity of a sector depends only on its own production, and is unrelated to the production linkages with other sectors, thus K is a diagonal matrix with diagonal elements greater than 0. The larger the diagonal elements, the stronger the recovery capacity of the corresponding sector in response to shocks. t is the discrete time variable, and $q(t)$ represents the sectoral inoperability vector at time t. Approximating Equation (2.9) to differential form:

$$\dot{q}(t) = K\left[A^* q(t) + c^*(t) - q(t)\right] \tag{2.10}$$

Solving Equation (2.10) we can get the equation describing the evolution of sectoral inoperability over time:

$$q(t) = e^{-K(I-A^*)t} q(0) + \int_0^t K e^{-K(I-A^*)(t-z)} c^*(z)\, dz \tag{2.11}$$

Assuming that the demand shock c^* remains unchanged, and $c^* = 0$, then Equation (2.11) can be simplified to:

$$q(t) = e^{-K(I-A^*)t} q(0) \tag{2.12}$$

Where $q(0)$ represents the initial inoperability vector of sectors after being shocked. As time goes by, the inoperability changes at a rate of $e^{-K(I-A^*)t}$.

2.1.4 Initial Inoperability

The initial inoperability $q(0)$ is calculated as:

$$q_i(0) = \underbrace{(1-\mathrm{RLI}_i)\,(1-\mathrm{ESS}_i)}_{\text{Workforce Unavailability}} \times \frac{\mathrm{LAPI}_i}{\text{Sector Output}_i} \tag{2.13}$$

The first factor, workforce unavailability, is adapted from [8, 9]. RLI_i is the Remote Labor Index, representing the probability that a worker in sector i can work from home. ESS_i is the Essential Score, representing the probability that a worker has an essential job that permits on-site work during a lockdown. The product $(1-\mathrm{RLI}_i)(1-\mathrm{ESS}_i)$ gives the expected fraction of workers who cannot work during the lockdown.

The second factor, $\mathrm{LAPI}_i/\text{Sector Output}_i$, captures how dependent each sector is on labour as an input, where LAPI is the Local Area Personal Income (equivalent to compensation of employees in the IO table).

2.1.5 Elasticity Coefficient

According to Equation (2.12), the inoperability of sector i at time t is:

$$q_i(t) = q_i(0)\, e^{-k_i(1-a_{ii}^*)t} \tag{2.14}$$

From Equation (2.14), we can derive the calculation formula for k_i:

$$k_i = \frac{\ln\left[q_i(0)/q_i(T)\right]}{T_i\,(1-a_{ii}^*)} \tag{2.15}$$

Where $q_i(0)$ is the initial inoperability of sector i; $q_i(T)$ is the inoperability of sector i at time T; a_{ii}^* is the element on the diagonal of matrix A^*, representing the self-dependence of sector i.

2.1.6 Economic Loss

Under the analytical framework of the DIIM model, the cumulative economic loss for sector i is:

$$EL_i = x_i \int_{t=0}^{t=T} q_i(t)\, dt \tag{2.16}$$

Where x_i is the planned output of sector i under normal production.

2.2 Key Sector Identification Methods

A key sector is defined as one where targeted intervention—modelled as a 10% reduction in initial inoperability $q(0)$—produces the largest reduction in total economic loss. Below we first describe the intervention and loss reduction evaluation procedure, followed by the expensive benchmark and five simplified structural alternatives.

2.2.1 Intervention and Loss Reduction Evaluation

Each method produces a ranking of sectors. To evaluate a method's effectiveness, we select its top-k ranked sectors and simulate a targeted intervention: the initial inoperability of each selected sector is reduced by 10%. The DIIM is then re-run with this modified $q(0)$ to obtain the post-intervention total economic loss. The difference between the baseline loss (no intervention) and the post-intervention loss gives the economic loss reduction achieved by that method's top-k selection. This procedure is applied identically to the DIIM benchmark and to every simplified method, so that all methods are compared on the same basis.

The formal procedure is given in Algorithm 1.

Algorithm 1: Intervention and Loss Reduction Evaluation

Input: Initial inoperability $q(0)$, interdependency matrix A^*, demand perturbation c^*, total output vector x, number of key sectors k, sector ranking method m

Output: Percentage loss reduction $\%\Delta EL_m$ for method m

```
// Step 1: Compute baseline economic loss (no intervention)
```

$\text{model}_{\text{base}} \leftarrow \text{DIIM}\big(q(0), A^*, c^*, x\big)$;

$EL_{\text{base}} \leftarrow \text{TotalEconomicLoss}\big(\text{model}_{\text{base}}\big)$;

```
// Step 2: Rank sectors using method m and select top-k
```

$\text{ranking} \leftarrow \textsc{RankSectors}(m)$;

$S_k \leftarrow \text{ranking}[1:k]$; // top-k sector indices

```
// Step 3: Apply intervention (reduce q(0) by 10% for selected
   sectors)
```

$q'(0) \leftarrow q(0)$;

for *each sector* $i \in S_k$ **do**

  $q'_i(0) \leftarrow 0.9 \times q_i(0)$;

```
// Step 4: Re-run DIIM with the modified initial inoperability
```

$\text{model}_{\text{int}} \leftarrow \text{DIIM}\big(q'(0), A^*, c^*, x\big)$;

$EL_{\text{int}} \leftarrow \text{TotalEconomicLoss}\big(\text{model}_{\text{int}}\big)$;

```
// Step 5: Compute loss reduction
```

$\Delta EL_m \leftarrow EL_{\text{base}} - EL_{\text{int}}$;

$\%\Delta EL_m \leftarrow \dfrac{\Delta EL_m}{EL_{\text{base}}} \times 100$;

return $\%\Delta EL_m$;

In words, the algorithm proceeds as follows. First, the DIIM is run with the original $q(0)$ to establish the baseline total economic loss, which represents the scenario with no policy intervention. Second, a sector ranking method m (either DIIM or a simplified method) produces an ordered list of sectors, and the top k are selected. Third, the initial inoperability of each selected sector is reduced by 10%, representing a targeted policy intervention that partially restores the sector's production capacity at time zero. Fourth, the DIIM is re-run with this modified initial condition $q'(0)$ to obtain the post-intervention total economic loss. Finally, the loss reduction is computed as the difference between the baseline and post-intervention losses. This procedure is repeated for every method m, with identical $q(0)$, k, and intervention magnitude, so that differences in loss reduction reflect only the quality of each method's sector ranking.

2.2.2 DIIM-Based Identification (Benchmark)

The DIIM identifies key sectors by running the full simulation without intervention, then ranking sectors by their cumulative economic loss EL_i from Equation (2.16). The top-k sectors are selected for intervention. This is the "gold standard" but requires knowledge of $q(0)$ and c^*.

2.2.3 Total Output (x_i)

This is the simplest benchmark method. Sectors are ranked simply by their total planned economic output x_i. This method ignores the network structure entirely and assumes that intervening in the largest economic sectors will yield the greatest reduction in loss.

2.2.4 PCA $\times x_i$

Instead of using A directly, this approach constructs the integrated input-output coefficient matrix:

$$H = A(I-A)^{-1} = A \cdot L \tag{2.17}$$

where $L = (I-A)^{-1}$ is the Leontief inverse. The matrix H captures both direct and indirect effects through the production network. PCA is applied to H, and a structural distance is computed for each sector based on its Euclidean distance from the origin in the space of the principal components:

$$d_i^{\text{PCA}} = \sqrt{\sum_{k=1}^{n_pcs} v_{ik}^2} \tag{2.18}$$

where v_{ik} is the loading of sector i on principal component k. To account for economic size, this structural distance is weighted by the sector's total output x_i:

$$\text{Score}_i^{\text{PCA}} = d_i^{\text{PCA}} \times x_i \tag{2.19}$$

Sectors with larger $\text{Score}_i^{\text{PCA}}$ are ranked higher.

2.2.5 PageRank $\times x_i$

The interdependency matrix A^* is treated as the weighted adjacency matrix of a directed graph. The PageRank algorithm [10] is applied to compute a structural centrality score PR_i for each sector, which measures importance based on the structure of incoming links with a damping factor. This centrality score is then weighted by the sector's total output:

$$\text{Score}_i^{\text{PR}} = PR_i \times x_i \tag{2.20}$$

2.2.6 Backward Linkage $\times x_i$ (BL $\times x_i$)

Backward linkages measure the extent to which a sector draws on other sectors' outputs as inputs [11, 12]. They are calculated as the column sums of the Leontief inverse:

$$BL_j = \sum_{i=1}^{n} l_{ij} \tag{2.21}$$

where l_{ij} are elements of L. A high BL_j indicates that sector j generates large demand for inputs from other sectors. To identify key sectors, this structural linkage is weighted by total output:

$$\text{Score}_j^{\text{BL}} = BL_j \times x_j \tag{2.22}$$

2.2.7 Forward Linkage $\times x_i$ (FL $\times x_i$)

Forward linkages measure how much a sector's output is used as inputs by other sectors [11, 13]. They are calculated as the row sums of the Leontief inverse:

$$FL_i = \sum_{j=1}^{n} l_{ij} \tag{2.23}$$

A high FL_i indicates that sector i is an important supplier to the rest of the economy. This structural linkage is weighted by total output:

$$\text{Score}_i^{\text{FL}} = FL_i \times x_i \tag{2.24}$$

2.3 Performance Metric

To compare a simplified method against DIIM, we define the *performance ratio*:

$$R_m = \frac{\Delta EL_m}{\Delta EL_{\text{DIIM}}} \tag{2.25}$$

where ΔEL_m is the total economic loss reduction achieved by intervening in the top-k sectors identified by simplified method m, and ΔEL_{DIIM} is the reduction achieved by the DIIM's top-k sectors. If $R_m \geq 0.95$, the simplified method is considered "close enough" to substitute for DIIM.

Chapter 3

Case Study 1: COVID-19 Pandemic

This chapter applies the DIIM framework to Singapore's COVID-19 pandemic, presenting the data, simulation results, and risk management analysis.

3.1 Data and Model Parameters

3.1.1 Input-Output Table

The IO transaction data for Singapore is obtained from the Department of Statistics, Singapore [14]. This study uses the 2019 IO table because it is the closest year before COVID-19 and represents the pre-pandemic economic structure. The available IO table is aggregated into 20 broad sectors, and this study further aggregates them into 15 sectors to match the Ministry of Manpower dataset used to compute initial inoperability. The mapping is provided in Appendix A (Table 8.1).

From the aggregated IO table, the technical coefficient matrix A is computed. A cross-section of A is provided in Appendix A (Table 8.2).

3.1.2 Initial Inoperability

Initial inoperability is calculated using Equation (2.13). The Remote Labor Index (RLI) for each sector is estimated from telecommuting rates reported by the Ministry of Manpower for 2020 [15]. Direct MOM data is available for Information & Communications and Financial Services (77%), Health & Social Services (28%), and Accommodation & Food Services (11%); remaining sectors are estimated from occupational mix. The Essential Score (ESS) is derived from the Ministry of Trade and Industry's list of essential services permitted to operate during the circuit breaker [16]. The resulting workforce unavailability ranges from 5% (Information & Communications) to 81% (Construction), and initial inoperability $q_i(0)$ ranges from 0.6% to 43.9%. The complete values are listed in Appendix A (Table 8.3).

3.1.3 Model Parameters

Singapore's lockdown measures ("circuit breaker") were implemented from 7 April to 1 June 2020, which is 55 days [4, 5]. This study sets the shock duration to 55 days.

In late April 2022 (751 days after the start of lockdown), Singapore's DORSCON level was lowered from Orange to Yellow [17], and employees were allowed to return to workplaces. This study assumes a final inoperability $q_i(T_i) = 1\%$, meaning economic activity returns to 99% of pre-lockdown levels after 751 days.

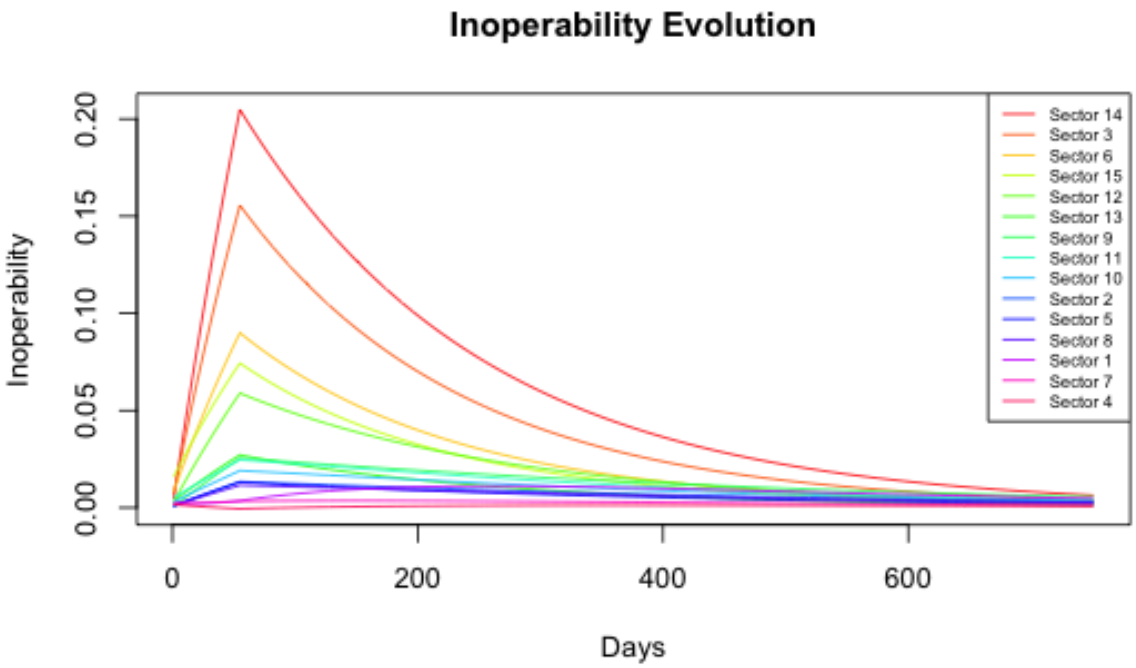

Figure 3.1: Inoperability evolution over the shock-and-recovery horizon (COVID-19, 15 sectors).

3.2 Results

3.2.1 Inoperability Evolution

Figure 3.1 shows the simulated inoperability trajectories for the 15 aggregated sectors over the shock-and-recovery horizon. Inoperability rises during the shock window as disruptions propagate through input-output linkages, and it declines during recovery as sectors gradually restore production. Because sectors are interconnected, each sector experiences both direct effects from its own disruption and indirect effects transmitted from other sectors.

3.2.2 Economic Loss Evolution

Figure 3.2 presents the corresponding economic-loss evolution. Sectors with larger pre-disruption outputs can accumulate large absolute losses even if their relative inoperability is not the highest, while smaller sectors may show high inoperability but smaller absolute losses.

3.2.3 Identifying Vulnerable Sectors

This section identifies vulnerable sectors using two measures: inoperability (relative production loss) and cumulative economic loss (absolute output loss). Figure 3.3 plots each sector using its rank in inoperability (y-axis) and its rank in economic loss (x-axis), so sectors that rank high in both measures are easy to identify.

To support prioritisation, we define ordinal "zones" (Top 5, Top 7, Top 10) as the intersection of sectors that appear in the top-k rankings of both measures. For example, the Top 5 zone contains sectors that are simultaneously ranked in the top five for both inoperability and economic loss, highlighting sectors that are both heavily disrupted and economically important.

Differences between the two rankings are expected. For example, a sector such as Arts, Entertainment & Recreation may rank high in inoperability but lower in economic loss if its pre-disruption output is smaller than that of other sectors. Overall, Figure 3.3 provides a practical view for decision-making when

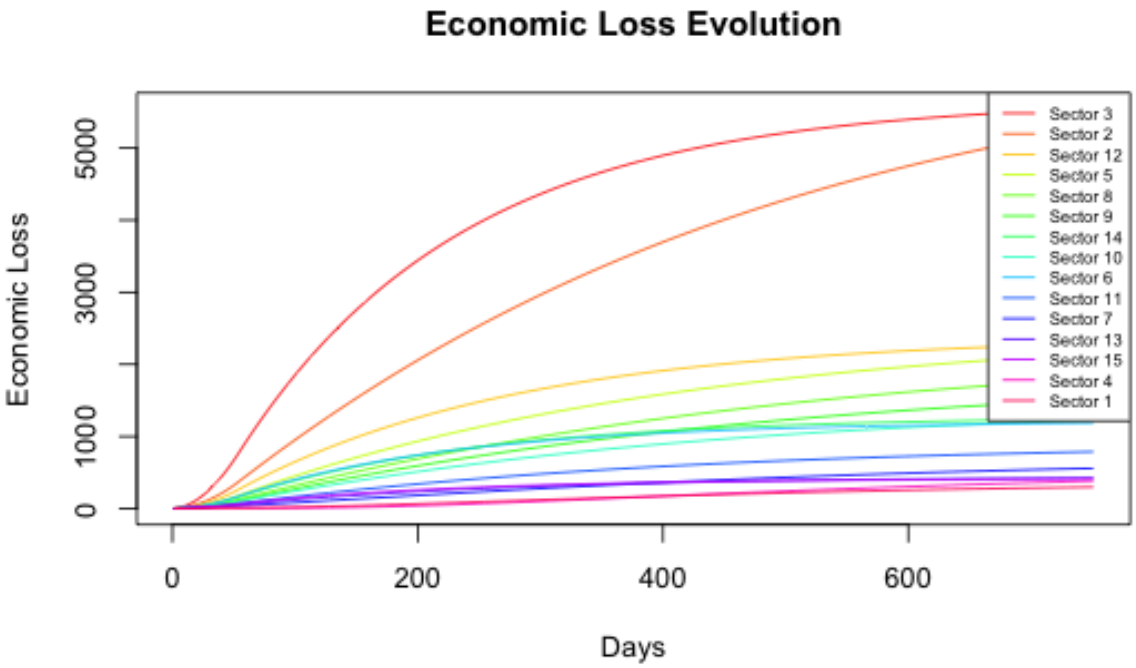

Figure 3.2: Economic loss evolution over the shock-and-recovery horizon (COVID-19, 15 sectors).

	2	3	5	12	8	9	10	14	6	11	7	13	15	4	1
14								X							
3		X													
6									X						
12				X											
15													X		
9						X									
13												X			
11										X					
2	X														
10							X								
5			X												
8					X										
1															X
7											X				
4														X	

Top 5 Zone	
3	Construction
12	Public Administration & Education

Top 7 Zone	
9	Real Estate Services

Top 10 Zone	
14	Arts, Entertainment & Recreation
6	Accommodation & Food Services
11	Administrative & Support Services
2	Manufacturing
10	Professional Services

Figure 3.3: Joint impact matrix: inoperability rank vs. economic loss rank (COVID-19).

policymakers want to balance restoring functionality (low inoperability) and reducing macroeconomic damage (low loss).

Chapter 4

Case Study 2: Manpower Disruption

This chapter extends the DIIM framework to a manpower disruption scenario motivated by Singapore's dependence on foreign labour and the potential for grey zone conflicts to trigger workforce departures.

4.1 Background: Singapore's Foreign Worker Dependence

Singapore's economy relies on approximately 1.4 million foreign workers, who constitute roughly one-third of the total labour force [3]. These workers are concentrated in sectors such as construction, manufacturing, marine shipyard, process industries, and domestic services. While this reliance on foreign labour has been a key enabler of economic growth, it creates a structural vulnerability to events that could trigger rapid workforce departures.

Grey zone conflicts—coercive actions that fall below the threshold of conventional warfare—represent a plausible threat scenario [6]. Such conflicts could include economic sanctions, cyber attacks on critical infrastructure, disinformation campaigns, or military posturing in the region. Any of these could erode confidence among foreign workers and their home governments, potentially triggering a significant and rapid outflow of non-resident workers.

Unlike the COVID-19 scenario, where disruption arose from both workforce unavailability ($q(0) > 0$) and reduced demand ($c^* > 0$), a manpower disruption is characterised by:

- **High initial inoperability** ($q(0) > 0$): Sectors lose productive capacity as workers depart.
- **No external demand shock** ($c^* = 0$): The disruption is purely structural. Demand may persist but cannot be met due to workforce shortages.

Because $c^* = 0$, the DIIM solution simplifies to $q(t) = e^{-K(I-A^*)t}\, q(0)$ when $c^* = 0$ (Equation (2.12)), meaning the disruption dynamics are governed entirely by the structure of the economy (A^*) and the initial shock ($q(0)$).

4.2 Data and Model Parameters

4.2.1 Input-Output Table

This scenario uses the 2022 Singapore IO table, aggregated to 15 broad sectors matching the same sectoral classification as the COVID-19 scenario. This aggregation facilitates direct comparison between the two case studies, while still capturing the heterogeneous distribution of foreign workers across sectors.

4.2.2 Initial Inoperability

Initial inoperability is computed using the same method as Equation (2.13), with workforce unavailability data reflecting the proportion of foreign workers in each sector. Sectors with high foreign worker concentrations (e.g., construction, process industries) have correspondingly higher initial inoperability.

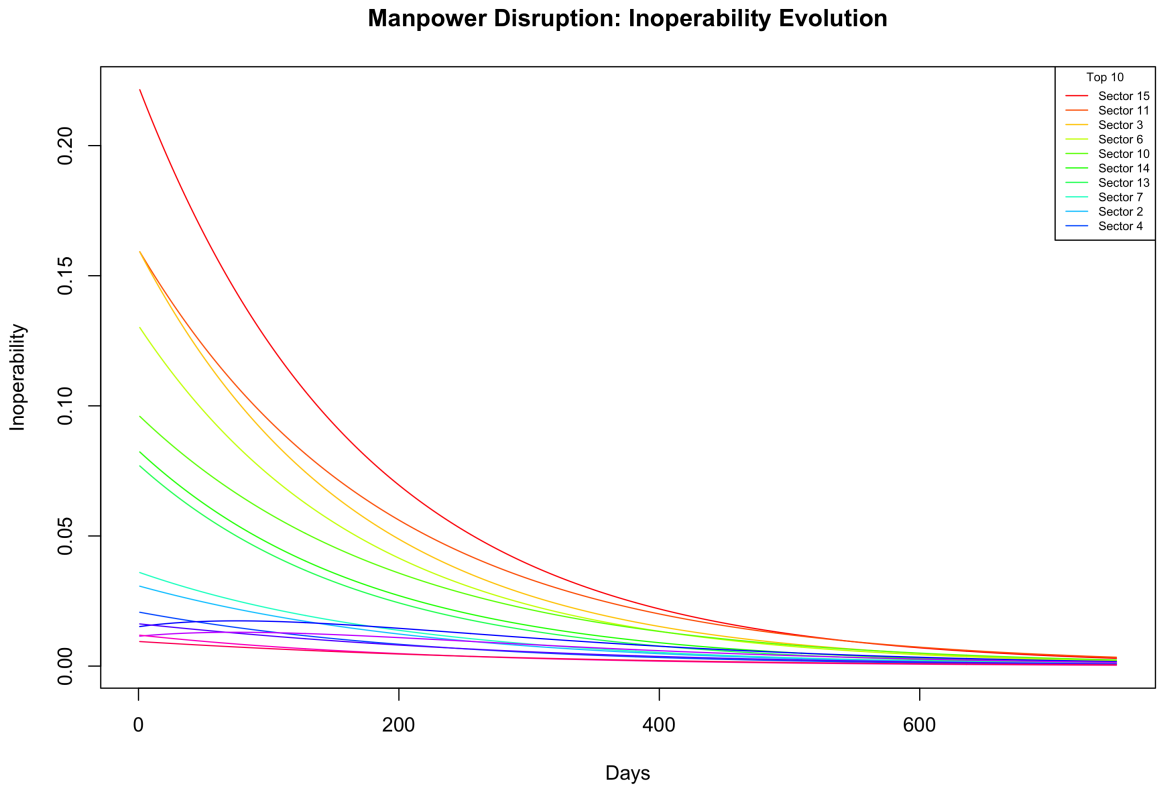

Figure 4.1: Inoperability evolution for the manpower disruption scenario (15 sectors).

4.2.3 Model Parameters

The shock duration is set to 55 days to enable comparison with the COVID-19 scenario, though a manpower disruption could have a different timeline. The total simulation horizon is 751 days with a final inoperability of 1%, representing near-full recovery through workforce replacement, retraining, or automation.

4.3 Results

4.3.1 Inoperability Evolution

Figure 4.1 shows the simulated inoperability trajectories for the 15 sectors. Compared to the COVID-19 scenario, the manpower disruption exhibits:

- Greater dispersion in initial inoperability across sectors, reflecting the uneven distribution of foreign workers.
- Purely structural cascading effects through the production network, as disruptions in labour-intensive sectors propagate to downstream industries without any external demand shock.

4.3.2 Economic Loss Evolution

Figure 4.2 presents the cumulative economic loss evolution. The 15-sector model reveals that even sectors with moderate inoperability can accumulate substantial losses if their pre-disruption output is large (e.g., manufacturing, financial services).

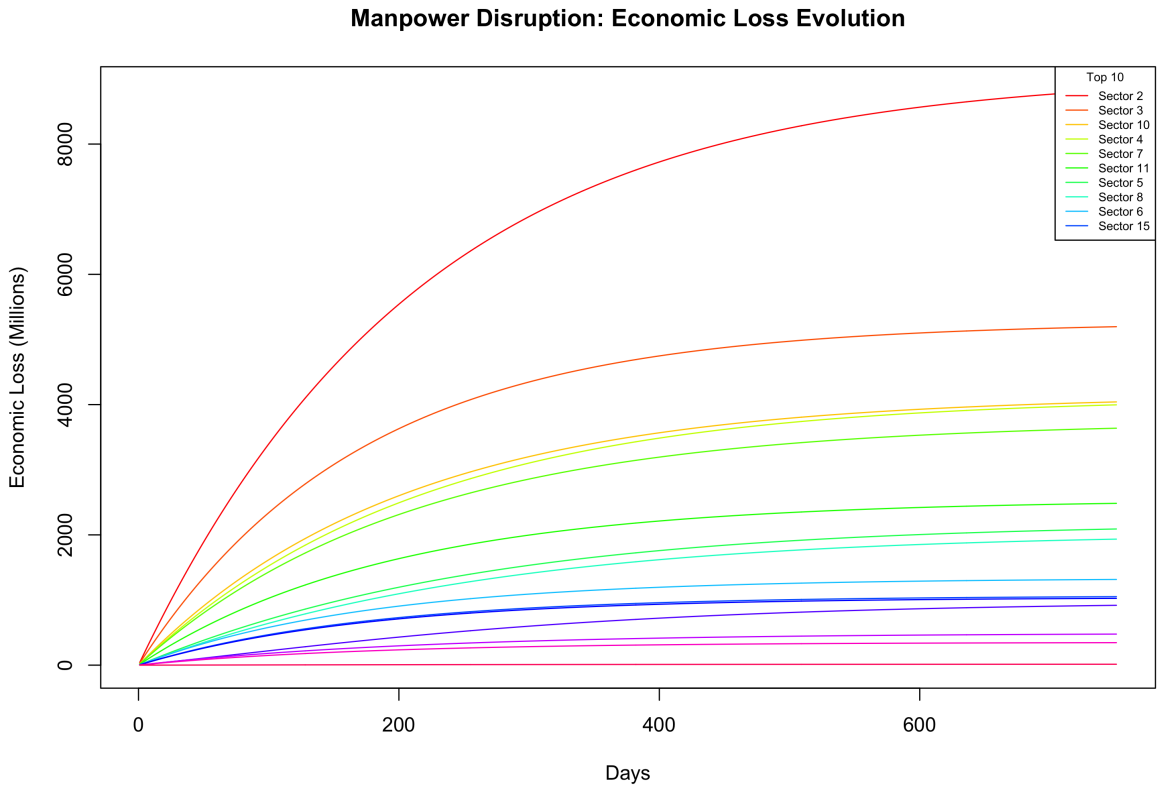

Figure 4.2: Economic loss evolution for the manpower disruption scenario (15 sectors).

4.4 Comparison with COVID-19

Table 4.1 summarises the key structural differences between the two scenarios.

Although both scenarios operate at the same 15-sector level, the manpower scenario is structurally distinct because the absence of a demand shock ($c^* = 0$) means the entire disruption propagation is driven by the production network alone. Furthermore, the 2022 IO table captures post-pandemic structural shifts in the economy. These structural differences affect the effectiveness of simplified key sector identification methods, as explored in the following chapters.

Parameter	**COVID-19**	**Manpower**
Number of sectors	15	15
IO table year	2019	2022
Demand shock c^*	> 0	$= 0$
Shock type	Demand + supply	Pure supply
Lockdown duration	55 days	55 days
Total horizon	751 days	751 days

Table 4.1: Comparison of COVID-19 and manpower disruption scenario parameters.

Chapter 5

Key Sector Identification Methods

This chapter compares the key sector identification methods described in Chapter 2 using the actual disruption data from both case studies. Recall that the DIIM identifies key sectors by running the full simulation and ranking by economic loss (expensive, requires $q(0)$ and c^*), while the simplified methods use only the IO table structure.

5.1 PCA Applied to the IO Matrix

PCA is applied to the integrated IO matrix $H = A(I-A)^{-1}$. Table 5.1 lists the PCA loadings for the COVID-19 scenario (15 sectors), sorted by Euclidean distance in the PC space, along with the final weighted score $\text{Score}_i^{\text{PCA}}$ which determines the ranking.

Sector ID	**PC1**	**PC2**	d_i^{PCA}	$\textbf{Score}_i^{\textbf{PCA}}$
2	0.489	0.548	0.734	262,791
5	0.231	0.602	0.645	110,877
4	0.164	0.198	0.257	55,084
8	0.174	0.320	0.364	49,794
3	0.773	−0.159	0.789	49,288
7	0.107	0.232	0.256	26,141
10	0.141	0.185	0.233	14,362
9	0.066	0.120	0.137	7,513
11	0.100	0.154	0.184	5,862
1	0.082	0.203	0.219	3,986
12	0.021	0.034	0.040	2,271
6	0.019	0.040	0.044	1,028
15	0.010	0.016	0.019	179
13	0.002	0.004	0.005	106
14	0.004	0.006	0.008	77

Table 5.1: PCA loadings, Euclidean distance, and weighted score $\text{Score}_i^{\text{PCA}}$ for COVID-19 sectors, sorted by weighted score.

The top-5 PCA $\times x_i$ sectors for COVID-19 are sectors 2 (Manufacturing), 5 (Transportation & Storage), 4 (Wholesale & Retail Trade), 8 (Financial & Insurance Services), and 3 (Construction). While sector 3 has the largest structural distance, sector 2 becomes the highest-ranked when weighted by its massive economic size.

For the manpower scenario (15 sectors), PCA is computed on the corresponding H matrix from the 2022 aggregated IO table. The top-5 PCA $\times x_i$ sectors are sectors 2, 5, 4, 7, and 8. These closely mirror

the COVID-19 PCA sectors, reflecting the similar aggregated structure, though the different IO table vintage (2022 vs 2019) and the absence of demand shock ($c^* = 0$) produce some ranking differences.

5.2 Simplified Method Rankings

Table 5.2 and Table 5.3 show the top-5 sectors identified by each method for the two scenarios.

Method	**Top-5 Sectors (COVID-19)**
Total Output	2, 4, 5, 8, 7
PCA $\times x_i$	2, 5, 4, 8, 3
PageRank $\times x_i$	2, 4, 5, 8, 7
BL $\times x_i$	2, 4, 5, 8, 7
FL $\times x_i$	2, 4, 5, 8, 7

Table 5.2: Top-5 sectors by each simplified method (COVID-19, 15 sectors).

Method	**Top-5 Sectors (Manpower)**
Total Output	2, 4, 5, 7, 8
PCA $\times x_i$	2, 5, 4, 7, 8
PageRank $\times x_i$	4, 2, 5, 7, 8
BL $\times x_i$	2, 5, 4, 7, 8
FL $\times x_i$	2, 5, 4, 7, 8

Table 5.3: Top-5 sectors by each simplified method (Manpower, 15 sectors).

Several observations are worth noting:

1. **High method agreement**: By incorporating total output x_i as a weight, the methods converge strongly. For COVID-19, Total Output, PageRank $\times x_i$, BL $\times x_i$, and FL $\times x_i$ produce the exact same top-5 ranking (2, 4, 5, 8, 7). PCA $\times x_i$ differs only slightly by including sector 3 instead of 7.

2. **Economic size dominates**: The fact that structurally distinct measures (centrality, backward linkage, and forward linkage) yield nearly identical rankings once weighted by x_i indicates that a sector's pre-disruption economic size is the dominant driver of its overall impact in these aggregated models.

3. **Cross-scenario consistency**: Despite using different IO tables (2019 vs 2022) and different shock structures ($c^* > 0$ vs $c^* = 0$), both scenarios produce very similar simplified method rankings. The manpower scenario rankings (2, 4/5, 5/4, 7, 8) closely mirror the COVID-19 rankings (2, 4, 5, 8, 7), confirming that economic size is the overwhelmingly dominant factor in these 15-sector models.

5.3 Comparison at Original Disruption Data

Using the actual $q(0)$ data from each scenario, we compare each simplified method's economic loss reduction against the DIIM benchmark when intervening in the top-5 sectors.

For the COVID-19 scenario, the DIIM identifies sectors 2, 3, 4, 5, 12 as key sectors and achieves a loss reduction of 3.91%. In contrast, Total Output, PageRank $\times x_i$, BL $\times x_i$, and FL $\times x_i$ achieve 2.65%, while PCA $\times x_i$ achieves 3.38%—a notable improvement over the purely size-based methods. The DIIM selects Construction (3) and Public Administration (12) in place of Financial Services (8) and Information

& Communications (7), reflecting its ability to capture sectors with high initial inoperability and strong structural dependencies.

For the manpower scenario, PCA $\times x_i$ matches the other simplified methods at 4.84%, while the DIIM achieves 6.88%—a substantially larger gap. This indicates that when disruptions are concentrated in structurally important sectors, the DIIM's ability to account for both $q(0)$ and c^* becomes more valuable.

This observation motivates the Monte Carlo analysis in Chapter 6: the performance of simplified methods at a single $q(0)$ point is insufficient to draw general conclusions. We need to understand how performance varies across the space of possible $q(0)$ vectors.

5.4 Effect of the Number of Key Sectors

Figure 5.1 shows how the economic loss reduction varies across all methods as the number of key sectors k increases, for the COVID-19 scenario. Across all methods, increasing k improves the loss reduction, but the rate of improvement differs. The simplest benchmark, Total Output, tracks very closely with the other structurally-weighted methods, confirming that weighting by economic size captures the bulk of the necessary information at this scale.

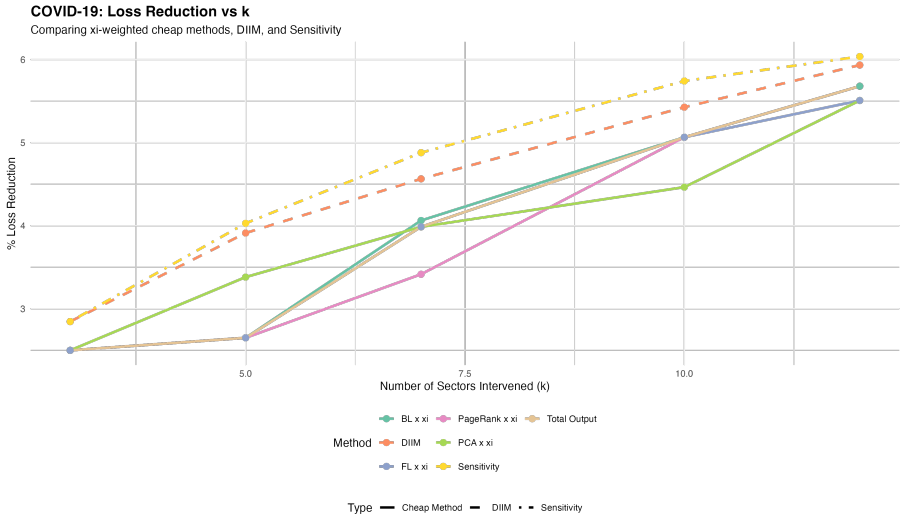

Figure 5.1: Economic loss reduction for different methods across top-k sectors (COVID-19).

Figures 5.2 and 5.3 compare the standard PCA-based and DIIM-based approaches specifically.

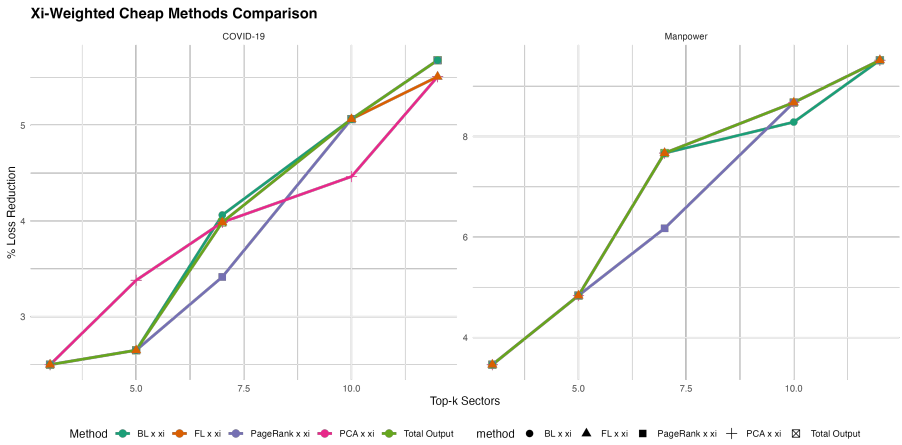

Figure 5.2: Economic loss reduction for PCA vs DIIM (COVID-19).

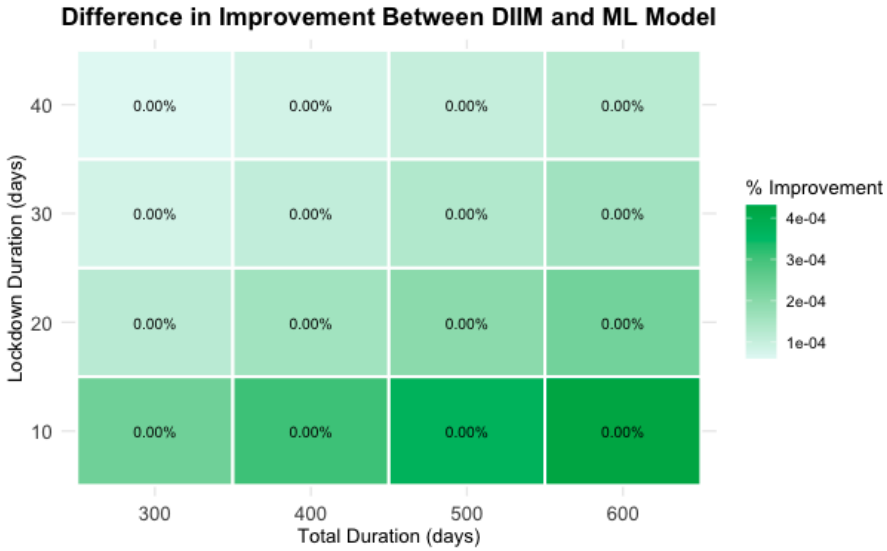

Figure 5.3: Difference in improvement: PCA vs DIIM (COVID-19).

These results show that PCA can perform well at the original COVID-19 data point, but the question remains: does this hold across a wider range of disruption profiles? This is addressed in the next chapter.

Chapter 6

Monte Carlo Robustness Analysis

The previous chapter showed that simplified methods can approach DIIM at specific $q(0)$ points but not universally. This chapter uses Monte Carlo simulation to systematically evaluate the robustness of simplified methods across a range of realistic disruption profiles and intervention scales.

6.1 Framework

6.1.1 Motivation

All simplified methods produce a *fixed* sector ranking—they depend only on the IO table structure (A, x), not on the disruption profile ($q(0)$). The DIIM, by contrast, adapts its ranking to the specific $q(0)$. A simplified method performs well when its fixed ranking happens to align with where $q(0)$ concentrates; it fails when the disruption hits sectors that the simplified method did not prioritise.

This insight suggests that the effectiveness of a simplified method is predictable from (i) features of $q(0)$ and (ii) the number of sectors targeted for intervention k. At small k, only the very top sectors matter, while at large k the specific ranking becomes less consequential because most sectors are covered regardless.

6.1.2 Performance Metric

For each simplified method m and a given $q(0)$, we compute the performance ratio from Equation (2.25). The mean ratio $\bar{R}_m$ across trials is the primary metric: it captures how much of the DIIM's loss reduction a simplified method achieves on average. As a secondary metric, we report the *close rate*—the fraction of trials where $R_m \geq 0.80$, i.e., the simplified method captures at least 80% of the DIIM's benefit.

6.2 Monte Carlo Design

6.2.1 Perturbation of Real $q(0)$

Rather than drawing $q(0)$ from a uniform distribution—which would generate many unrealistic disruption profiles—we perturb the actual scenario-specific $q(0)$ vector using log-normal noise:

$$q_i(0)^{\text{trial}} = q_i(0)^{\text{actual}} \cdot \exp(\varepsilon_i), \quad \varepsilon_i \sim \mathcal{N}(0,\, 0.5^2) \tag{6.1}$$

with the constraint $q_i(0)^{\text{trial}} \in [10^{-6},\, 1]$. This preserves the realistic structure of the disruption (sectors that were heavily impacted remain relatively more impacted) while introducing meaningful variation. The log-normal noise ensures that the perturbation is multiplicative—a sector with $q(0) = 0.01$ can plausibly move to 0.005 or 0.02, but not jump to 0.5.

6.2.2 Simulation Procedure

For each scenario and each value of $k \in \{3, 5, 7, 10, 12\}$, we draw 2 000 perturbed $q(0)$ vectors. For each trial:

1. Perturb $q(0)^{\text{actual}}$ using Equation (6.1).
2. Run the DIIM without intervention and compute total economic loss.
3. Run the DIIM with intervention in the DIIM's top-k sectors (10% $q(0)$ reduction) and compute ΔEL_{DIIM}.
4. For each simplified method m, run the DIIM with intervention in method m's top-k sectors and compute ΔEL_m.
5. Compute $R_m = \Delta EL_m / \Delta EL_{\text{DIIM}}$ and record whether $R_m \geq 0.80$ ("close").

This yields 20 000 simulation trials per scenario (2 000 $\times$ 5 values of k), for a total of 40 000 trials across both scenarios.

6.3 Results: Performance Across k

6.3.1 Mean Performance Ratio

Tables 6.1 and 6.2 present $\bar{R}_m$ for each simplified method across all values of k.

Method	k=3	k=5	k=7	k=10	k=12
Total Output	0.870	0.641	0.818	0.908	0.953
PCA $\times x_i$	0.870	**0.817**	0.818	0.800	0.923
PageRank $\times x_i$	0.870	0.641	0.703	0.908	0.953
BL $\times x_i$	0.870	0.641	**0.834**	0.908	0.953
FL $\times x_i$	0.870	0.641	0.818	0.908	0.923

Table 6.1: $\bar{R}_m$ (simplified / DIIM) by k for COVID-19 (2 000 trials per k). Bold indicates the best simplified method at that k.

Method	k=3	k=5	k=7	k=10	k=12
Total Output	0.652	0.682	0.927	0.926	0.976
PCA $\times x_i$	0.652	0.682	0.927	0.926	0.976
PageRank $\times x_i$	0.652	0.682	0.749	0.926	0.976
BL $\times x_i$	0.652	0.682	0.927	0.885	0.976
FL $\times x_i$	0.652	0.682	0.927	0.926	0.976

Table 6.2: $\bar{R}_m$ (simplified / DIIM) by k for Manpower (2 000 trials per k).

6.3.2 Close Rates

Tables 6.3 and 6.4 present the close rates (fraction of trials where $R_m \geq 0.80$).

Method	k=3	k=5	k=7	k=10	k=12
Total Output	66.1%	5.7%	62.0%	96.7%	100.0%
PCA $\times x_i$	66.1%	**64.1%**	62.0%	53.4%	99.8%
PageRank $\times x_i$	66.1%	5.7%	17.7%	96.7%	100.0%
BL $\times x_i$	66.1%	5.7%	**67.2%**	96.7%	100.0%
FL $\times x_i$	66.1%	5.7%	62.0%	96.7%	99.8%

Table 6.3: Close rates ($R_m \geq 0.80$) by k for COVID-19 (2 000 trials per k). Bold indicates notable structural method advantage.

Method	k=3	k=5	k=7	k=10	k=12
Total Output	16.4%	17.2%	97.1%	99.2%	100.0%
PCA $\times x_i$	16.4%	17.2%	97.1%	99.2%	100.0%
PageRank $\times x_i$	16.4%	17.2%	33.6%	99.2%	100.0%
BL $\times x_i$	16.4%	17.2%	97.1%	93.5%	100.0%
FL $\times x_i$	16.4%	17.2%	97.1%	99.2%	100.0%

Table 6.4: Close rates ($R_m \geq 0.80$) by k for Manpower (2 000 trials per k).

6.3.3 Three Regimes of k

The results show a non-monotonic pattern that can be decomposed into three regimes:

1. **Small** k ($k = 3$): All simplified methods agree on the top 3 sectors and produce identical performance with $\bar{R}_m = 0.870$ (COVID-19) and 0.652 (Manpower). The dominant economic sectors (Manufacturing, Wholesale & Retail, Transportation & Storage) are so large that every method—whether based on size, linkage, or centrality—selects them. Two-thirds of trials achieve ≥80% of DIIM in COVID-19 (66.1%), but only 16.4% in Manpower. Structural distinctions between simplified methods are irrelevant at this scale.

2. **Medium** k ($k = 5$–7): The methods diverge, and structural information adds substantial value. At $k = 5$ in COVID-19, PCA $\times x_i$ achieves $\bar{R}_m = 0.817$ versus 0.641 for Total Output—a 27% relative improvement—and a close rate of 64.1% versus just 5.7%. This dramatic gap arises because PCA selects sector 3 (Construction) over sector 7 (Information & Communications). At $k = 7$, BL $\times x_i$ leads with $\bar{R}_m = 0.834$ and close rate of 67.2%, capturing sector 12 (Public Administration) via its backward linkage signal. This is the regime where the "marginal" sectors (positions 4–7 in the ranking) matter most, and each method's structural lens produces meaningfully different selections.

3. **Large** k ($k = 10$–12): The methods converge again as the intervention covers most of the economy. At $k = 12$ (80% of 15 sectors), close rates reach 100% for both scenarios, with $\bar{R}_m$ of 0.92–0.98. When intervening on nearly all sectors, the only question is which 3 sectors to *exclude*, and even suboptimal exclusions have limited impact.

6.4 Key Observations

The multi-k tables show several patterns:

1. **PCA $\times x_i$ outperforms Total Output at $k = 5$ (COVID-19)**: PCA $\times x_i$ achieves $\bar{R}_m = 0.817$ versus 0.641 for Total Output—a 27% relative improvement—and a close rate of 64.1% versus just 5.7%. This stems from PCA's inclusion of sector 3 (Construction), which has the highest structural

distance ($d_3^{\text{PCA}} = 0.789$) despite ranking only 6th in economic size. Construction's high initial inoperability ($q_3(0) = 0.221$) makes intervening there disproportionately effective.

2. **BL $\times x_i$ leads at $k = 7$ (COVID-19)**: BL $\times x_i$ achieves $\bar{R}_m = 0.834$ and a close rate of 67.2%, outperforming all other simplified methods including PCA $\times x_i$ (0.818, 62.0%). Its backward linkage signal captures sector 12 (Public Administration), whose strong input dependencies amplify the cascading effect of intervention. This shows that different structural methods have complementary strengths at different intervention scales.

3. **PageRank $\times x_i$ underperforms at medium k**: At $k = 7$, PageRank $\times x_i$ drops to $\bar{R}_m = 0.703$ (COVID-19) and 0.749 (Manpower), with close rates of just 17.7% and 33.6% respectively—far below the other methods. PageRank's random-walk centrality prioritises sector 12 over sectors 3 and 10 at positions 6–7, which proves suboptimal under perturbed $q(0)$.

4. **All methods converge at large k**: At $k = 10$–12, close rates reach 97–100% for most methods, with $\bar{R}_m$ of 0.91–0.98. The specific ranking matters little when the majority of sectors are covered.

5. **No single simplified method dominates across all k**: PCA $\times x_i$ is best at $k = 5$, BL $\times x_i$ is best at $k = 7$, and Total Output matches the best at $k = 3$, 10, and 12. This suggests that the optimal simplified method depends on the intervention scale.

6.5 Practical Implications

The multi-k analysis leads to the following practical guidelines:

1. **Large-scale intervention ($k \geq 10$)**: Simplified methods are reliable substitutes for DIIM. With $\bar{R}_m$ above 0.90 and close rates of 97–100%, the additional computational cost of DIIM is difficult to justify when intervening on the majority of sectors. Total Output alone—requiring no IO table analysis—achieves 95% of DIIM's performance at $k = 12$.

2. **Targeted intervention ($k = 3$–7)**: DIIM provides substantial value over purely size-based methods. At $k = 5$, Total Output achieves only 64% of DIIM's loss reduction (close rate 5.7%). However, structural methods significantly narrow this gap: PCA $\times x_i$ achieves 82% of DIIM (close rate 64.1%), and BL $\times x_i$ leads at $k = 7$ with 83% of DIIM (close rate 67.2%). While a gap to DIIM remains, structural methods provide a meaningful improvement over the naive size-based benchmark in this regime.

3. **Pre-event planning**: When $q(0)$ is unknown, simplified methods remain valuable for pre-positioning resources. The Total Output ranking identifies the sectors whose disruption would cause the greatest absolute economic loss, while PCA $\times x_i$ and BL $\times x_i$ can flag structurally important sectors that size alone would miss (e.g., Construction in the COVID-19 scenario).

4. **Crisis response**: Once $q(0)$ is observed, run the full DIIM if targeting fewer than 10 sectors. The computational cost of the simulation is modest (seconds on modern hardware) and the performance gain is substantial.

Chapter 7

Discussion

7.1 The Three Regimes of Intervention Scale

The main finding of the Monte Carlo analysis is that the relative performance of simplified methods is not constant—it depends critically on the number of sectors targeted for intervention. This "three regimes" pattern holds across both scenarios:

- **Small** k: All methods agree on the dominant sectors, so structural distinctions are irrelevant. Performance is moderate (ratios of 0.65–0.87) because even agreement on the top sectors does not guarantee alignment with the DIIM's adaptive ranking.
- **Medium** k: Methods diverge on the marginal sectors (positions 4–7), and structural information adds value. PCA $\times x_i$ gains 27% over Total Output at $k = 5$ in COVID-19; BL $\times x_i$ leads at $k = 7$. This is the regime where the question "which specific sectors?" matters most.
- **Large** k: The methods converge again as the intervention covers most of the economy. At $k = 12$, close rates reach 100% and $\bar{R}_m$ exceeds 0.95, making simplified methods reliable substitutes for the full DIIM simulation.

This pattern has a straightforward mathematical explanation. With 15 sectors and $k = 12$, there are only $\binom{15}{12} = 455$ possible intervention sets. Most reasonable rankings will select similar sets, because the question reduces to which 3 sectors to *exclude*. At $k = 5$, there are $\binom{15}{5} = 3{,}003$ possible sets, and the overlap between methods' selections is much smaller.

7.2 Why Economic Size Dominates Structure

A related finding is that weighting structural methods by a sector's pre-disruption economic size (x_i) overwhelmingly dominates the results. This can be understood through two mechanisms:

1. **The anchor of scale**: Pure structural measures (like unweighted centrality or linkage) can highlight highly interconnected sectors that are economically minuscule. Multiplying by total output ensures that structurally important but economically small sectors are not over-ranked. The massive variation in sector sizes acts as a strong anchor, pulling structurally distinct methods toward nearly identical rankings.
2. **The Total Output benchmark**: The pure Total Output benchmark (ranking sectors solely by x_i, completely ignoring the IO network) performs identically to the best structurally-weighted methods at most values of k. At $k = 3$, $k = 10$, and $k = 12$, Total Output matches or exceeds all structural methods. This suggests that for macro-level loss reduction, the initial scale of the intervention matters more than optimising for downstream cascading effects.

The exception is the medium-k regime, where structural signals can identify "hidden" key sectors. Sector 3 (Construction) in the COVID-19 scenario is the clearest example: despite ranking 6th in economic size, it has the highest PCA structural distance (0.789) and high initial inoperability ($q(0) = 0.221$). PCA $\times x_i$ captures this; Total Output does not.

7.3 When Do Structural Methods Add Value?

The results suggest that structural methods (PCA $\times x_i$, BL $\times x_i$) add value over the Total Output benchmark under specific conditions:

1. **Medium intervention scale**: The marginal benefit of structural information is concentrated at $k = 5$–7, where the "which specific sectors?" question is most consequential.

2. **Sectors with high structure–size mismatch**: When a sector has strong IO linkages or unusual propagation patterns but moderate economic size, structural methods can identify it while size-based methods miss it. This is scenario-dependent—the COVID-19 economy has such a sector (Construction), while the manpower economy does not, leading to identical simplified method performance in the latter.

3. **Concentrated disruptions**: Simplified methods perform best when $q(0)$ is concentrated in the sectors that the method already ranks highly. Structural methods are most valuable when the disruption pattern partially but imperfectly aligns with the economic size ranking—i.e., when some structurally important sectors are heavily disrupted but would not be prioritised by size alone.

7.4 Cross-Scenario Comparison

The two scenarios differ in an important respect: COVID-19 has both supply disruption ($q(0) > 0$) and demand shock ($c^* > 0$), while manpower has only supply disruption ($c^* = 0$).

At $k = 5$, the COVID-19 scenario shows differentiation among simplified methods (PCA $\times x_i$ at 0.817 vs others at 0.641), while the manpower scenario shows none (all at 0.682). This suggests that the presence of a demand shock (c^*) creates more heterogeneity in the DIIM's optimal ranking, which in turn gives structural methods more room to capture patterns that size alone misses.

At $k = 12$, both scenarios achieve close rates of 100%, confirming that at large intervention scales the specific ranking is immaterial. The convergence is faster in the manpower scenario: at $k = 7$, close rates already reach 97% (Manpower) versus 62% (COVID-19). This occurs because the pure supply-shock dynamics ($c^* = 0$) are more predictable—without a demand shock, the cascading pattern is fully determined by A^* and the initial $q(0)$, and covering roughly half the sectors is sufficient to approximate the DIIM's optimal intervention.

7.5 National Security Implications

From a national security perspective, the results support a two-stage approach to economic resilience planning:

1. **Peacetime (pre-event)**: Use Total Output to identify the economically dominant sectors for resilience investment. Supplement with PCA $\times x_i$ and BL $\times x_i$ to flag structurally critical sectors that may not be obvious from size alone (e.g., Construction in the COVID-19 scenario). This requires only the IO table—no disruption data.

2. **Crisis response**: Once $q(0)$ is observed, the appropriate tool depends on the intervention scale:

- If targeting $\leq$7 sectors: run the full DIIM. The performance gap is 18–36%, and the computational cost is modest.
- If targeting $\geq$10 sectors: simplified methods achieve $\geq$90% of DIIM's performance. Use Total Output for immediate prioritisation.

For Singapore specifically, the analysis suggests that sectors with high foreign worker concentration *and* strong IO linkages (e.g., Construction, Manufacturing) should be priority targets for workforce resilience measures. Even crude structural analysis using only the IO table can identify these sectors with reasonable accuracy.

7.6 Limitations

This study has several limitations:

1. **Uniform intervention**: All methods are compared using a standardised 10% reduction in $q(0)$ for top-k sectors. In practice, the feasible intervention magnitude may vary across sectors.
2. **Sector aggregation**: Both scenarios use 15-sector aggregations. At finer disaggregation, structural signals may become more discriminating and the advantage of structural methods over Total Output could increase.
3. **IO table vintage**: The COVID-19 scenario uses the 2019 IO table and the manpower scenario uses the 2022 IO table. Structural changes between these years are not controlled for.
4. **Recovery dynamics**: The DIIM assumes exponential recovery with fixed resilience coefficients. In practice, recovery may be non-exponential, sector-specific, and dependent on policy interventions.
5. $q(0)$ **perturbation**: While the log-normal perturbation around actual $q(0)$ is more realistic than uniform random draws, it still assumes that the disruption structure is anchored to the historical scenario. Entirely novel disruption patterns (e.g., a pandemic hitting different sectors) are not tested.

7.7 Future Work

Future extensions could include:

- Testing at finer sector disaggregation (e.g., 50+ sectors) where structural signals may be more discriminating.
- Incorporating sector-correlated disruptions in the Monte Carlo analysis.
- Extending the framework to multi-period disruptions where $q(0)$ evolves over time.
- Validating the decision rules on actual disruption data from other economies.
- Exploring variable intervention magnitudes across sectors, rather than a uniform 10% reduction.

Chapter 8

Conclusion

This project developed and applied a quantitative framework for assessing economic resilience under disruption, relevant to Singapore's national security planning.

Using the Dynamic Inoperability Input-Output Model (DIIM), two disruption scenarios were studied. The COVID-19 pandemic scenario (15 sectors, 2019 IO table) demonstrated how lockdown-induced workforce unavailability cascaded through input-output linkages to produce sector-level inoperability paths and cumulative economic losses. The manpower disruption scenario (15 sectors, 2022 IO table), motivated by the risk of grey zone conflicts triggering foreign worker departures, revealed how purely structural disruption ($c^* = 0$) produces different cascading dynamics compared to a combined supply-and-demand shock.

This study's main contribution is the systematic comparison of five simplified methods for identifying key sectors against the DIIM benchmark, evaluated across multiple intervention scales ($k = 3, 5, 7, 10, 12$) using 40 000 Monte Carlo trials with realistic $q(0)$ perturbations. The analysis reveals three distinct regimes:

- At **small** k ($k = 3$), all simplified methods agree on the dominant sectors and perform identically, achieving $\bar{R}_m$ of 0.65–0.87.
- At **medium** k ($k = 5$–7), methods diverge and structural information adds substantial value. PCA $\times x_i$ outperforms the Total Output benchmark by up to 27% in the COVID-19 scenario (close rate 64% vs 6%) by identifying structurally critical sectors (e.g., Construction) that size-based rankings miss. BL $\times x_i$ provides complementary value at $k = 7$ (close rate 67%). At $k = 5$, the best simplified method recovers only 82% of the loss reduction that DIIM achieves on average. However, structural methods close much of the gap relative to the Total Output benchmark.
- At **large** k ($k \geq 10$), simplified methods converge to DIIM performance, with $\bar{R}_m$ of 0.91–0.98 and close rates of 97–100%. At this scale, even the simplest method—ranking by Total Output alone—is a reliable substitute for the full simulation.

The near-equivalence between Total Output and structurally-weighted methods at most k values shows that a sector's absolute economic scale is the dominant predictor of macro-level loss reduction. Structural analysis adds value only in the medium-k regime, where the marginal sectors differ across methods, and only when the economy contains sectors with a significant mismatch between economic size and structural importance.

Simplified methods perform best when the disruption profile aligns with the economic size distribution—a condition that is scenario-dependent and cannot be assumed in advance.

For national security planning, this enables a scale-dependent approach: use simplified structural analysis for peacetime resilience investments and for large-scale crisis response ($k \geq 10$), but run the full DIIM when targeting fewer than 10 sectors during a crisis. The computational cost of the DIIM is modest, and the performance gain at small k is substantial.

The framework is generalisable beyond the two scenarios studied here. Any economy with an available IO table can compute the simplified structural rankings in peacetime and apply the multi-k analysis to determine the appropriate tool for its intervention scale. Future work could extend the approach to finer sector disaggregation, correlated disruptions, and multi-period dynamics.

Bibliography

[1] Joost R. Santos and Yacov Y. Haimes. Modeling the demand reduction input-output (i-o) inoperability due to terrorism of interconnected infrastructures. *Risk Analysis*, 24(6):1437–1451, 2004.

[2] Jian Jin and Haoran Zhou. A demand-side inoperability input–output model for strategic risk management: Insight from the COVID-19 outbreak in shanghai, china. *Sustainability*, 15(5):4003, 2023.

[3] Ministry of Manpower, Singapore. Foreign workforce numbers. https://www.mom.gov.sg/documents-and-publications/foreign-workforce-numbers, 2023. Accessed 2026-02-17.

[4] Ministry of Health, Singapore. End of circuit breaker, phased approach to resuming activities safely. https://www.moh.gov.sg/newsroom/end-of-circuit-breaker-phased-approach-to-resuming-activities-safely/, May 2020. Newsroom release dated 19 May 2020. Accessed 2026-02-17.

[5] Ministry of Health, Singapore. Post circuit breaker measures. https://www.moh.gov.sg/newsroom/post-circuit-breaker-measures/, 2020. Newsroom / Parliamentary reply page referencing circuit breaker dates. Accessed 2026-02-17.

[6] Ministry of Defence, Singapore. Factsheet: Singapore's total defence. https://www.totaldefence.gov.sg/, 2020. Total Defence overview including economic, digital, and military defence pillars. Accessed 2026-03-23.

[7] Wassily W. Leontief. Quantitative input and output relations in the economic system of the United States. *The Review of Economics and Statistics*, 18(3):105–125, 1936.

[8] R. Maria del Rio-Chanona, Penny Mealy, Anton Pichler, François Lafond, and J. Doyne Farmer. Supply and demand shocks in the COVID-19 pandemic: An industry and occupation perspective. *Oxford Review of Economic Policy*, 36(S1):S94–S137, 2020.

[9] Anton Pichler, Marco Pangallo, R. Maria del Rio-Chanona, François Lafond, and J. Doyne Farmer. In and out of lockdown: Propagation of supply and demand shocks in a dynamic input-output model. *Journal of Economic Dynamics and Control*, 143:104527, 2022.

[10] Sergey Brin and Lawrence Page. The anatomy of a large-scale hypertextual web search engine. In *Proceedings of the 7th International World Wide Web Conference*, pages 107–117, 1998.

[11] P. N. Rasmussen. Studies in inter-sectoral relations. 1956.

[12] Hollis B. Chenery and Tsunehiko Watanabe. Interdependence and industrial structure. *Econometrica*, 26(4):487–521, 1958.

[13] Erik Dietzenbacher. Interregional multipliers and backward and forward linkages. *Economic Systems Research*, 14(2):155–166, 2002.

[14] SINGSTAT (Singapore Department of Statistics). Industry by industry input-output table (supply, use and input-output tables 2019). https://data.gov.sg/datasets/d_959da3e78690cb3136c529fbc448d178/view. data.gov.sg dataset. Accessed 2026-02-17.

[15] Ministry of Manpower, Singapore. Labour force in Singapore 2020. https://stats.mom.gov.sg/Pages/Labour-Force-In-Singapore-2020.aspx, 2020. Table 52: Employed Residents Who Worked From Home by Industry. Accessed 2026-03-22.

[16] Ministry of Trade and Industry, Singapore. List of essential services. https://www.mti.gov.sg/, April 2020. Adjustments to essential services during circuit breaker, 7 April 2020. Accessed 2026-03-22.

[17] Ministry of Health, Singapore. Easing our COVID-19 measures. https://www.moh.gov.sg/newsroom/easing-our-covid-19-measures/, April 2022. DORSCON level lowered from Orange to Yellow effective 26 April 2022. Accessed 2026-03-23.

Appendix A: COVID-19 Scenario Data

Sector ID	PC1	PC2
3	0.772976374	-0.158540904
2	0.488575487	0.547564001
5	0.230631604	0.602425551
8	0.173952844	0.31984955
4	0.16376151	0.197548491
10	0.141148665	0.185477914
7	0.107051197	0.232037887
11	0.100445788	0.154054165
1	0.082357584	0.203204452
9	0.065932369	0.120308858
12	0.02097817	0.033819003
6	0.01904942	0.04009044
15	0.010160651	0.015616709
14	0.004471868	0.006256627
13	0.002201633	0.00402146

Table 8.1: Sector PCA loadings (COVID-19, 15 sectors)

	Agriculture, Fishing, Quarrying, Utilities and Sewerage & Waste Management	Manufacturing	Construction
Agriculture, Fishing, Quarrying, Utilities and Sewerage & Waste Management	4,308.828700	2,985.482100	198.954100
Manufacturing	498.915600	59,777.284000	4,939.792300
Construction	166.973900	498.152500	23,583.027800

Table 8.2: Cross-section of technical coefficient matrix (COVID-19)

Sector ID	Sector	Initial inoperability
1	Agriculture, Fishing, Quarrying, Utilities and Sewerage & Waste Management	0.020592
2	Manufacturing	0.033293
3	Construction	0.155924
4	Wholesale & Retail Trade	0.061522
5	Transportation & Storage	0.032185
6	Accommodation & Food Services	0.184162
7	Information & Communications	0.006077
8	Financial & Insurance Services	0.009252
9	Real Estate Services	0.026593
10	Professional Services	0.105436
11	Administrative & Support Services	0.123510
12	Public Administration & Education	0.115083
13	Health & Social Services	0.100575
14	Arts, Entertainment & Recreation	0.177582
15	Other Community, Social & Personal Services	0.439401

Table 8.3: Sector Initial Inoperability (COVID-19)

Appendix B: Manpower Scenario Data

The manpower disruption scenario uses the 2022 Singapore IO table with 15 sectors. Due to the large number of sectors, only summary statistics are presented here.

Parameter	**COVID-19**	**Manpower**
Number of sectors (n)	15	15
IO table year	2019	2022
Mean initial inoperability	0.10608	varies
Max initial inoperability	0.43940	varies
Demand shock (c^*)	> 0	$= 0$
Lockdown duration	55 days	55 days
Total horizon	751 days	751 days
PCA $\times x_i$ top-5 sectors	2, 5, 4, 8, 3	2, 5, 4, 7, 8
DIIM top-5 sectors	by simulation	by simulation

Table 8.4: Summary of scenario parameters

Appendix C: Code

The R code for all simulations and analyses is available in the project repository. Key scripts include:

- `functions.R`: Core DIIM function, data loading, and simplified method implementations (Total Output, PCA $\times x_i$, PageRank $\times x_i$, BL $\times x_i$, FL $\times x_i$)
- `pca.R`: PCA analysis on the integrated IO matrix
- `simulations/decision_rules.R`: Monte Carlo framework, logistic regression, and decision boundary rules
- `simulations/enhanced_comparison.R`: Multi-PC PCA, varying top-k sweep, and benchmark vs structured method comparison
- `simulations/run_simulations.R`: Parameter sweeps for severity (c^*) and lockdown duration impacts
- `manpower-disruption/manpower-disruption.R`: Manpower scenario DIIM simulation implementation